



Universidad Autónoma de Querétaro
Facultad de Informática
Base de Datos II



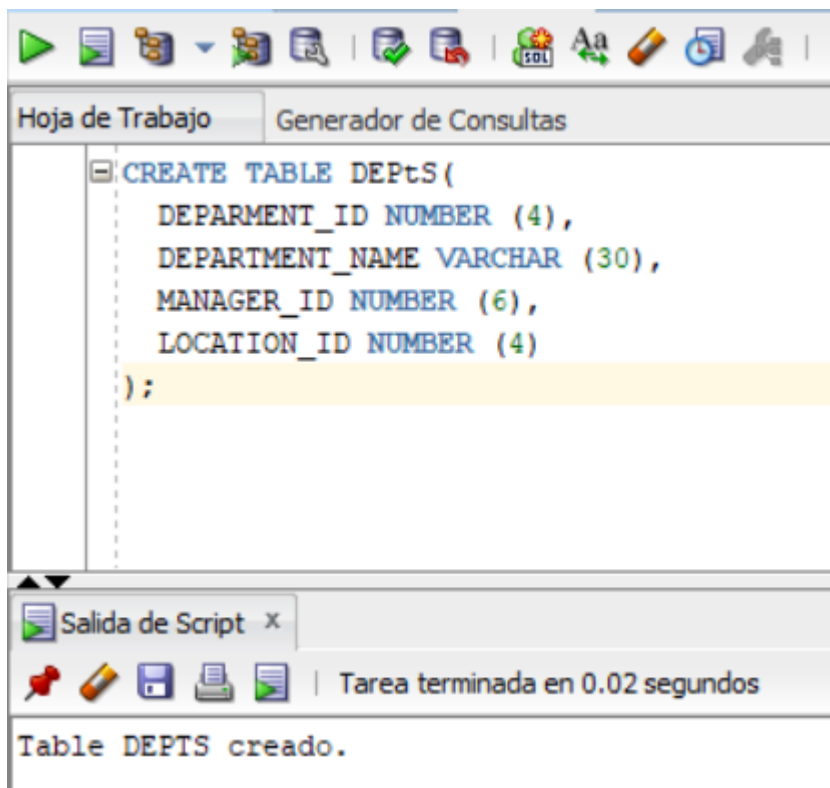
Práctica #15. Secuencias, Índices y Sinónimos en Oracle 11g

Instrucciones: Contesta los siguientes ejercicios. Adjunta, para cada uno, el comando SQL que lo resuelva y una captura de pantalla del resultado.

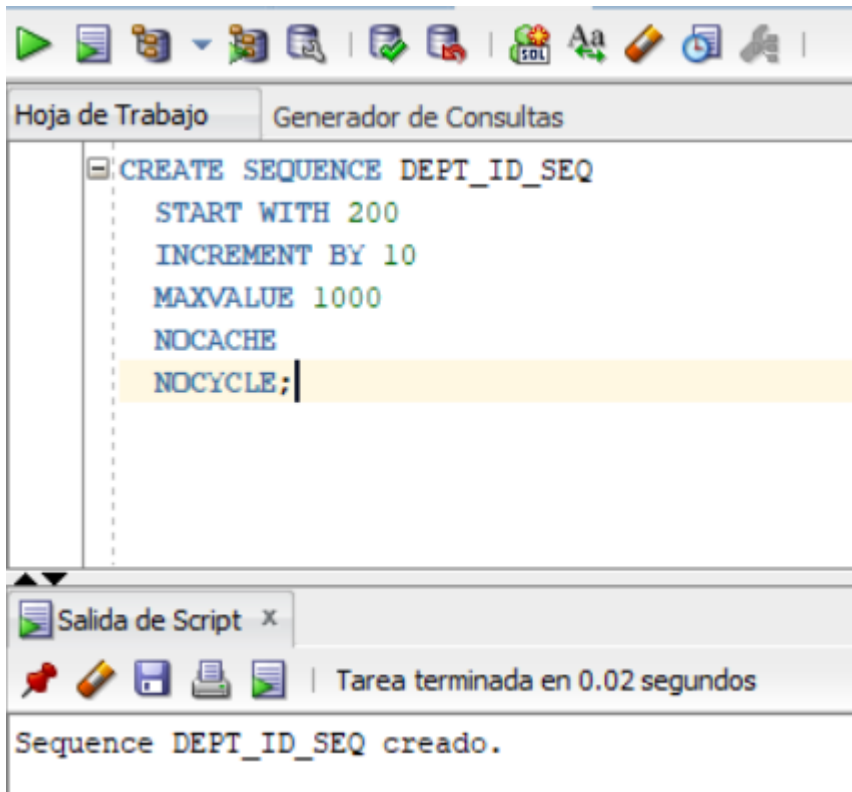
AWR AWR AWR

1. Crear la tabla DEPTS, como una copia de la estructura de la tabla DEPARTMENTS. Mantén limpia la nueva tabla, para usarla posteriormente.

```
CREATE TABLE DEPTS(  
    DEPARTMENT_ID NUMBER (4),  
    DEPARTMENT_NAME VARCHAR (30),  
    MANAGER_ID NUMBER (6),  
    LOCATION_ID NUMBER (4)  
);
```



2. Cree una secuencia para utilizarla con la columna de clave primaria de la tabla DEPTS. La secuencia debe comenzar en 200 y tener un valor máximo de 1000. Haga que la secuencia aumente sus valores de diez en diez. Asigne a la secuencia el nombre DEPT_ID_SEQ.



3. Escriba la siguiente consulta:

```
SELECT * FROM all_sequences WHERE sequence_name = 'DEPT_ID_SEQ';
```

¿Qué resultado obtienes?

The screenshot shows the 'Resultado de la Consulta' (Query Result) window. It displays the results of the query 'SELECT * FROM all_sequences WHERE sequence_name = 'DEPT_ID_SEQ';'. The results are shown in a table with the following columns: SEQUENCE_OWNER, SEQUENCE_NAME, MIN_VALUE, MAX_VALUE, INCREMENT_BY, CYCLE_FLAG, ORDER_FLAG, CACHE_SIZE, and LAST_NUMBER. The results are as follows:

SEQUENCE_OWNER	SEQUENCE_NAME	MIN_VALUE	MAX_VALUE	INCREMENT_BY	CYCLE_FLAG	ORDER_FLAG	CACHE_SIZE	LAST_NUMBER
1 HR	DEPT_ID_SEQ	1	1000	10 N	N		0	200

4. Inserte dos filas en la tabla DEPTS. Asegúrese de utilizar la secuencia que creó para la columna ID. Agregue dos departamentos llamados Educación y Administración. Confirme las adiciones.

```
INSERT INTO DEPTs (department_id, department_name)  
VALUES (DEPT_ID_SEQ.NEXTVAL, 'Educacion');
```

```
INSERT INTO DEPTs (department_id, department_name)  
VALUES (DEPT_ID_SEQ.NEXTVAL, 'Administracion');
```

```
INSERT INTO DEPTS (department_id, department_name)
VALUES (DEPT_ID_SEQ.NEXTVAL, 'Educación');

INSERT INTO DEPTS (department_id, department_name)
VALUES (DEPT_ID_SEQ.NEXTVAL, 'Administración');
```

Script Output x

Task completed in 0.089 seconds

1 row inserted.

1 row inserted.

5. Ejecute las siguientes consultas:

```
CREATE TABLE EMP AS SELECT * FROM EMPLOYEES;
```

```
ALTER TABLE EMP
ADD CONSTRAINT fk_emp_dept
FOREIGN KEY (department_id)
REFERENCES departments(department_id);
```

console_2 [hr] console_3 [hr] x

Tx: Manual ✓ ↺ Playground v

```
1 ✓ CREATE TABLE EMP AS SELECT * FROM EMPLOYEES;
2
3 ✓ ALTER TABLE EMP
4     ADD CONSTRAINT fk_emp_dept
5     FOREIGN KEY (department_id)
6     REFERENCES departments(department_id);
7
```

[2026-03-04 12:22:03] HR> CREATE TABLE EMP AS SELECT * FROM EMPLOYEES
[2026-03-04 12:22:04] 107 rows affected in 155 ms
[2026-03-04 12:22:04] HR> ALTER TABLE EMP
ADD CONSTRAINT fk_emp_dept
FOREIGN KEY (department_id)
REFERENCES departments(department_id)
[2026-03-04 12:22:04] completed in 109 ms

6. Visualice a todos los empleados de la tabla EMP que laboren en los departamentos del 50 en adelante. Verifique el tiempo de ejecución de la consulta.

```
SELECT EMPLOYEE_ID FROM EMP
WHERE DEPARTMENT_ID > 50;
```

Output HR.EMP

FOREIGN KEY (department_id)
REFERENCES departments(department_id)

[2026-03-04 12:22:04] completed in 109 ms

[2026-03-04 12:30:37] HR> SELECT EMPLOYEE_ID FROM EMP
WHERE DEPARTMENT_ID > 50

[2026-03-04 12:30:38] 51 rows retrieved starting from 1 in 921 ms (execution: 16 ms, fetching

7. Cree un índice en la columna de llave foránea (DEPARTMENT_ID) en la tabla EMP.

```
CREATE INDEX EMP_DEPT_ID_IDX ON EMP(department_id);
```

Output HR.EMP

[2026-03-04 12:22:04] completed in 109 ms

[2026-03-04 12:30:37] HR> SELECT EMPLOYEE_ID FROM EMP
WHERE DEPARTMENT_ID > 50

[2026-03-04 12:30:38] 51 rows retrieved starting from 1 in 921 ms (execution: 16 ms, fetching

[2026-03-04 12:32:20] HR> CREATE INDEX EMP_DEPT_ID_IDX ON EMP(department_id)

[2026-03-04 12:32:20] completed in 18 ms

8. Ejecute la instrucción del ejercicio 6. Verifique el cambio en el tiempo de ejecución de la consulta.

```
6 REFERENCES departments(department_id);
7
8
9 SELECT EMPLOYEE_ID FROM EMP
10 WHERE DEPARTMENT_ID > 50;
11
12 CREATE INDEX EMP_DEPT_ID_IDX ON EMP(department_id);
13
14 ✓ SELECT * FROM EMP WHERE department_id >= 50;
```

Output HR.EMP

WHERE DEPARTMENT_ID > 50

[2026-03-04 12:30:38] 51 rows retrieved starting from 1 in 921 ms (execution: 16 ms, fetching: 905 ms)

[2026-03-04 12:32:20] HR> CREATE INDEX EMP_DEPT_ID_IDX ON EMP(department_id)

[2026-03-04 12:32:20] completed in 18 ms

[2026-03-04 12:33:15] HR> SELECT * FROM EMP WHERE department_id >= 50

[2026-03-04 12:33:16] 96 rows retrieved starting from 1 in 593 ms (execution: 20 ms, fetching: 573 ms)

9. Visualice los índices que existe en el diccionario de datos para la tabla EMP. Utilice la vista USER_INDEXES.

```
CREATE TABLE EMP AS SELECT * FROM EMPLOYEES;

ALTER TABLE EMP
ADD CONSTRAINT fk_emp_dept
FOREIGN KEY (department_id)
REFERENCES departments (department_id);
```

Salida de Script x Resultado de la Consulta x

10. Borra la secuencia y el índice utilizados en la pregunta 2 y 7. Verifica que ya NO existan dichos objetos consultando las vistas USER_SEQUENCES y USER_INDEXES.

The screenshot shows the SQL Developer interface. The top pane contains the following SQL script:

```
DROP SEQUENCE DEPT_ID_SEQ;  
DROP INDEX EMP_DEPTID_INDEX;
```

The bottom pane shows the execution results:

Sequence DEPT_ID_SEQ borrado.
Index EMP_DEPTID_INDEX borrado.

Below this, a new query is executed to check the status of the objects:

```
SELECT * FROM USER_SEQUENCES WHERE SEQUENCE_NAME = 'DEPT_ID_SEQ';
```

The result shows 0 rows, confirming the sequence has been dropped.

Another query is executed to check the index:

```
SELECT * FROM USER_INDEXES WHERE INDEX_NAME = 'EMP_DEPTID_INDEX';
```

The result shows 0 rows, confirming the index has been dropped.

11. Cree un sinónimo EMPLEA, para la tabla de EMPLOYEES y confirma la creación del sinónimo visualizando la tabla USER_SYNONYMS.

The screenshot shows the SQL Developer interface. The top pane contains the following SQL script:

```
CREATE SYNONYM EMPLEA FOR EMPLOYEES;  
SELECT * FROM USER_SYNONYMS;
```

The bottom pane shows the execution results:

CREATE SYNONYM EMPLEA FOR EMPLOYEES;
SELECT * FROM USER_SYNONYMS;

The result shows 1 row, confirming the synonym has been created:

SYNONYM_NAME	TABLE_OWNER	TABLE_NAME	DB_LINK
EMPLEA	HR	EMPLOYEES	(null)

12. Realizar una consulta de todos los empleados, contratados entre el año 2000 y 2003, usando el sinónimo anterior.

The screenshot shows the SQL Developer interface. The top pane contains the following SQL script:

```
SELECT employee_id, first_name, last_name, hire_date  
FROM EMPLEA  
WHERE hire_date BETWEEN DATE '2000-01-01' AND DATE '2003-01-01';
```

The bottom pane shows the execution results:

Resultado de la Consulta x

Todas las Filas Recuperadas: 8 en 0.044 segundos

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	HIRE_DATE
102	Lex	De Haan	13/01/01
108	Nancy	Greenberg	17/08/02
109	Daniel	Faviet	16/08/02
114	Den	Raphaely	07/12/02
203	Susan	Mavris	07/06/02
204	Hermann	Baer	07/06/02
205	Shelley	Higgins	07/06/02
206	William	Gietz	07/06/02